

Prisms and tilting equivalence

1) Perfectoid rings

Last time: constructed for S perfect

$$i) \underbrace{W(S)}_{\substack{\text{Hilbert's} \\ \text{assect.}}} \left[\text{Witt vectors of } S \right]$$

S is p -complete, p -torsion free.

$$\cdot \frac{W(S)}{pW(S)} \xrightarrow{\sim} S$$

$$ii) R^b = \varprojlim_{(-)^p} R_{pR} \xrightarrow{\sim} \varprojlim_{(-)^p} R$$

$$\text{with } \# : R^b \rightarrow R$$

$$(\bar{x}_n) \mapsto \lim_{n \rightarrow \infty} (x_n)^{p^n}$$

Cor S perfect.

Teichmüller lift

$$\exists! (-)^\flat : S \rightarrow W(S) \text{ mult section}$$

of $W(S) \rightarrow S$ projection.

$$\text{Proof } (W(S))^b = \left(\frac{W(S)}{pW(S)} \right)^b = S^b = S$$

So

$$\begin{array}{ccc} S & & \\ \exists! \downarrow & \searrow = & \text{From previous prop.} \\ W(S) & \xrightarrow{\#} & S \end{array} \quad \square$$

Cor S perfect. $x \in W(S)$. Then

$$\exists! ([x_i]) \in W(S)^{\mathbb{N}}$$

$$x = \sum_{i=0}^{\infty} [x_i] p^i$$

$$\text{Proof } W(S) \xrightarrow{\text{mod } p} S$$

$$x \mapsto x_0$$

$$[x_0] \longleftarrow$$

$$\text{So } x - [x_0] = py \text{ unique by } p\text{-tors free}$$

Continue with $y = y - [y_0]$, and

similarly inductively $\square$

Proposition (Fontaine's θ -map)

R p -complete. Then $\exists!$ ring map

$$\begin{array}{ccc} \theta : W(R^b) & \longrightarrow & R \\ \uparrow (-)^\flat & \nearrow \# & \\ R^b & & \end{array} \quad \text{commutes}$$

Proof Have to be:

$$\theta \left(\sum [x_i] p^i \right) \mapsto \sum \#(x_i) p^i$$

Can show "by hand" that it

is a ring morphism.

"Simpler"

Deformation theory: As R^b perfect

$$\forall n \quad W(R^b)_{p^n W(R^b)} \longleftarrow \mathbb{Z}_p_{p^n \mathbb{Z}_p}$$

"formally etale"

$$\Rightarrow \exists! \text{ lift } R^b \rightarrow R_{pR}$$

as a ring map. Has to be

the above $\square$

Def (Distinguished element) S perfect

$d \in W(S)$ is "distinguished" if

$$S(d) \in W(S)^\times$$

$$\text{Eg: } p \in W(S); (x_0, x_1, \dots) \in W(S) \quad x_1 \in S^\times$$

Def (Perfectoid ring) R a ring

is "perfectoid" if

i) p -complete

$$ii) \exists \varpi \in R \quad \varpi^p = p\alpha \quad \alpha \in R^\times$$

(and $\varpi \in \text{Im}(R^b \rightarrow R)$)

$$iii) W(R^b) \xrightarrow{\theta} R \text{ and}$$

$$\ker(\theta) = (d)$$

$\nwarrow$ distinguished.

$\square$

Remark $d = [\varpi^p] + p\alpha$ in the form

of the cor. before.

$\rightarrow$ Can take ϖ with comp.

Seq of p -power roots.

Prop (Equiv def) R a ring is

pfid if i) and ii) and

$$iii)' (-)^p : R_{\varpi R} \xrightarrow{\sim} R_{pR}$$

$$(-)^p : R[\varpi] \xrightarrow{\sim} R[p]$$

Proof Omitted in full.

Observe however:

$$\begin{array}{ccc} R^b_{\varpi^b} & \xrightarrow{(A)} & R_{\varpi} \\ \downarrow (-)^p & \wr & \downarrow (-)^p \\ R^b_{\varpi^b p} & \xrightarrow{(B)} & R_{pR} \end{array}$$

$$\text{If } W(R^b) \xrightarrow{(-)^p} R$$

iso then

modulo $[\varpi^b]$ or

$[\varpi^b p]$ gives (A)

and (B).

$\square$

Exercise R p -tors free ϖ such

that $\varpi^p = p\alpha$.

$$R_{\varpi} \xrightarrow{(-)^p} R_p \iff R \hookrightarrow R\left[\frac{1}{p}\right]$$

p -int closed.

Examples a) Any perfect ring is.

$$1) \mathbb{Z}_p \langle t^{1/p^\infty} \rangle = \left(\frac{\mathbb{Z}_p[t^{1/p^\infty}]}{(t-p)} \right)^{\wedge p}$$

So modulo p :

$$\mathbb{F}_p[t^{1/p^\infty}]$$

$$\xrightarrow{t}$$

$$\varpi = t^{1/p}$$

works.

Tilt: $\mathbb{F}_p \langle t^{1/p^\infty} \rangle$ (t -completed)

2) Closure of int clac domain

(exercise).

3) R pfid. $R \langle t^{1/p^\infty} \rangle$

$$R_{\varpi} \langle t^{1/p^\infty} \rangle \xrightarrow{(-)^p} R_p \langle t^{1/p^\infty} \rangle \text{ still holds.}$$

$$4) \mathbb{Z}_p^{\text{cyc}} = \mathbb{Z}_p \langle \zeta_{p^\infty} \rangle = \mathbb{Z}_p \langle t^{1/p^\infty} \rangle$$

(See exercises)

$$\xrightarrow{(t^{p-1} + \dots + 1)}$$

2) Prisms, perfect prisms

Def A prism is a pair (A, I) where A is (p, I) -complete ^(derived) where A is a δ -ring and I is a Cartier divisor which is locally generated by a distinguished element. $\hat{}$ (faithfully flat)

E.g. 1) $(W(S), (p))$ (crystalline) $A \rightarrow A' \in \delta\text{-ring } IA' = d \cdot A' \cdot \text{unit}$

2) R perfectoid.

Notation $W(R^b) := A_{\text{inf}}(R)$.

Then $(A_{\text{inf}}(R), \ker \theta)$ is a prism.

3) K perfect field.

$(W(K)[[u]], E(u))$
 $\varphi: u \mapsto u^p$ $\hat{}$ Eisenstein pol for some uniformizer.

E.g. $(\mathbb{Z}_p[[t]], t^n - p)$ $n \in \mathbb{N}$
 $t \mapsto t^{p^n}$

Lemma (Rigidity of prisms)

$(A, I) \rightarrow (B, J)$ map of prisms.

Then $B \otimes_A I \xrightarrow{\sim} J$ in part $IB = J$.

Proof For simplicity say $I = (d)$

$J = (e)$. (Gen case by faith. flat descent)

So $d = ef$ for some f .

So $\delta(d) = e^p \delta(f) + f^p \delta(e)$
 $\text{unit} \quad \quad \quad \text{unit}$

by (p, e) -completeness ^(derived)

$f^p \text{ unit} \Rightarrow f \text{ unit} \quad \square$

Def A prism is called "perfect" if φ_A is an iso.

Proposition R pfd, $(d) = \ker \theta$.

$\{\text{Perfect prisms}\} \xrightarrow[\sim]{\text{mod } d, A_{\text{inf}}(R)} \{\text{pfd } R\text{-algebras}\}$

Proof Fix $R \rightarrow S$ pfd alg.

Then $A_{\text{inf}}(S) \xrightarrow[d]{} S$ iso

by rigidity of prisms.

Say $A_{\text{inf}}(R) \rightarrow P$ perfect prism.

$A_{\text{inf}}((P/d)^b) \rightarrow P$

iso modulo P by

$(P/d)^b = (P/(P, d))^b$

$\hat{} = (P/P)^{d^b} \hat{} = P/P$

See exercises.

P/P is d -complete

as P is (p, d) complete.

Corollary R pfd ring. $\square$

$\{R^b - \omega^b\text{-complete algs}\}$

$\uparrow \quad \uparrow (-)^b$

$\{R \text{ pfd-algebras}\}$

Proof

$\{R^b - \omega^b\text{-complete algs}\} \xrightarrow[\sim]{\text{mod } d, A_{\text{inf}}(R) \text{ perf prisms}}$

$\{R \text{ pfd-algebras}\} \xrightarrow[\sim]{\text{mod } d}$

$\square$

